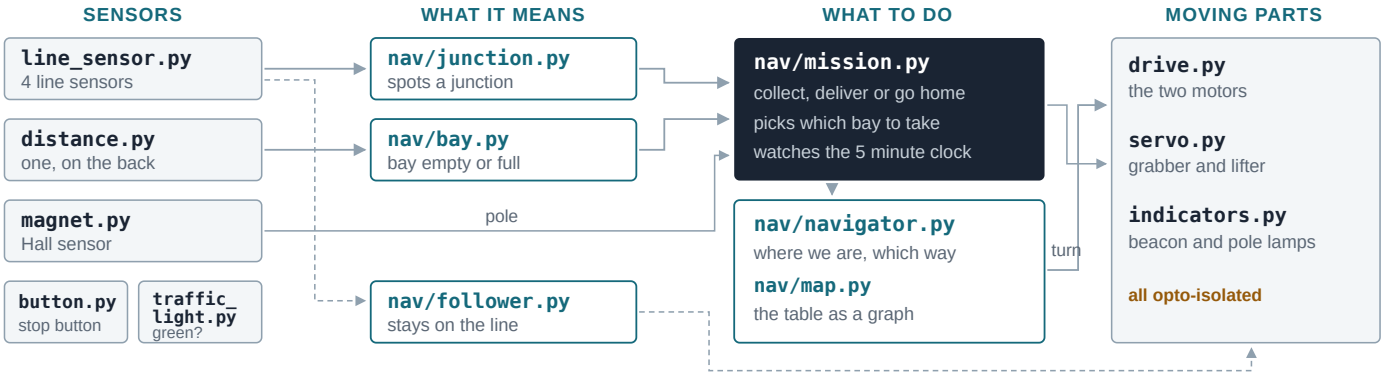
PicoBot AGV — Code Structure and Algorithms

IDP Group M108 · Software subteam · IDP M1 2026 · AGV software in `sw/` (V3.5), branch `rear_bay_detect`

The AGV **counts junction markings** instead of measuring distance, so the same code runs on all four tables. It **turns round inside a shelf bay** to check with its rear sensor whether the bay is free. Where a package goes is decided by two things: the **side** it was collected from, and its **magnet pole**.

1 HOW THE CODE IS STRUCTURED

`main.py` runs the loop every 5 ms — reads the sensors, asks nav/ what to do, drives the motors



`config.py` every pin number and every tuned value, kept in one place

Dashed arrows are line following, which runs when nothing else applies. `drivers/` talks to the pins; `nav/` makes the decisions.

2 THE MAIN FUNCTIONS

Function	Gives back	What it does
`JunctionDetector.poll()`	the junction type	One event per junction
`BayDetector.poll()`	empty / full	Reads the rear sensor, parked in a bay
`Mission.bay_to_check()`	a bay, or nothing	The next bay worth checking
`Mission.bay_checked(full)`	—	Remembers a full bay for the rest of the run
`Mission.collected(pole)`	—	Plans the delivery
`State.bays_beside()`	the bay on each side	Which bays are here, from the map
`State.junction_turn()`	left / right / straight	The turn to make next
`map.route(start, end)`	a list of junctions	The shortest way there

3 THE MAIN LOOP, CHECKED IN THIS ORDER EVERY 5 MS

1	**Stop button pressed?**	**yes** → **stop everything and end the run**
2	**Flash the amber beacon**	On and off twice a second
3	**At a delivery bay?**	**yes** → **let the box go**, reverse out and turn
4	**At a junction?**	**yes** → at a pickup bay, **grab the box and read its pole**. At a shelf bay worth checking, **run the check below**. Otherwise turn
5	**Red traffic light ahead?**	**yes** → wait at the foot of the ramp until it goes green
6	**None of these**	Follow the line

Collecting → ***delivering*** → ***returning***: *the AGV works the pickup bays on one side, delivers each box, and goes home when time runs short. Nothing in the loop waits. Turns and grabs are timed inside the loop, so the stop button is still checked all the way through. The motors are cut on every exit, including an error.*

Checking a shelf bay

There is one distance sensor, on the back. To use it the AGV turns round in the bay:

1 Turn into the bay, front first.
2 Drive in until just short of the shelf.
3 U-turn, so the back faces the shelf.
4 Square up, stop, and measure.
5 **Empty** — turn back and drop the box. **Full** — drive out and try the next bay.

Every full bay is remembered and driven past after that.

Where a package goes

Picked up	Pole	Shelf
Right side	north / south	RU / RB
Left side	north / south	LU / LB

The side is free — the AGV drove there, so it knows. The pole is read three times before it is believed.

With no rear sensor fitted the check is skipped, and the bays are filled 1, 2, 3 in order.

Generative AI declaration: all algorithmic flows were devised by the group. AI was used for repetitive changes, formatting, debugging and commenting.
Repository: `github.com/blakehrobinson-design/IDP-repo/tree/rear_bay_detect/V3.5/sw`

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